Exno: 11
Date: 1/10/2026

Simulation of RIP using cisco Packet tracer

Aim: To Simulate Routing information Protocol (RIP) using cisco packet tracer and observe dynamic routing between different networks.

IP configuration:

Device	Interface	IP Configuration
PC0	FastEthernet	10.0.0.2/8
Router 0	Fa0/1	10.0.0.1/8
Router 0	S0/0/1	192.168.1.253/30
Router 0	S0/0/0	192.168.1.249/30
Router 1	S0/0/0	192.168.1.250/30
Router 1	S0/0/1	192.168.1.246/30
Router 2	S0/0/0	192.168.1.245/30
Router 2	S0/0/1	192.168.1.253/30
Router 2	Fa0/1	20.0.0.1/30
PC1	Fast Ethernet	20.0.0.2/30

Procedure:

1) Create the topology with 3 routers and 2 PCs
2) Connect the routers using serial cables and connect PC's to the respective routers
3) Assign IP Address to all PC's and router Interface.

4) Enable the router interfaces.

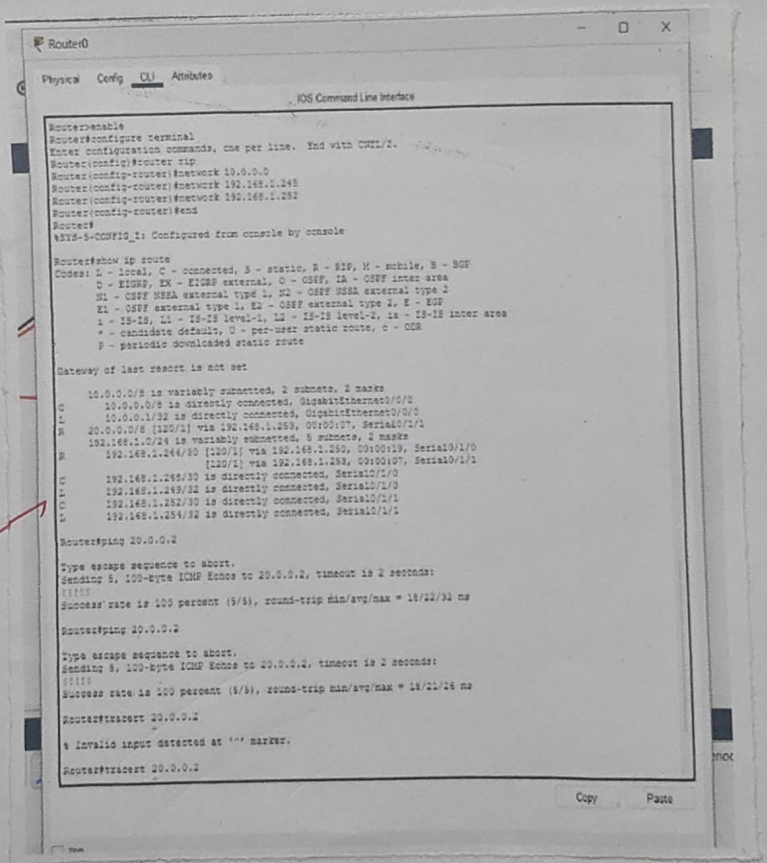
5) Configure the remaining interfaces according to the given IP table.

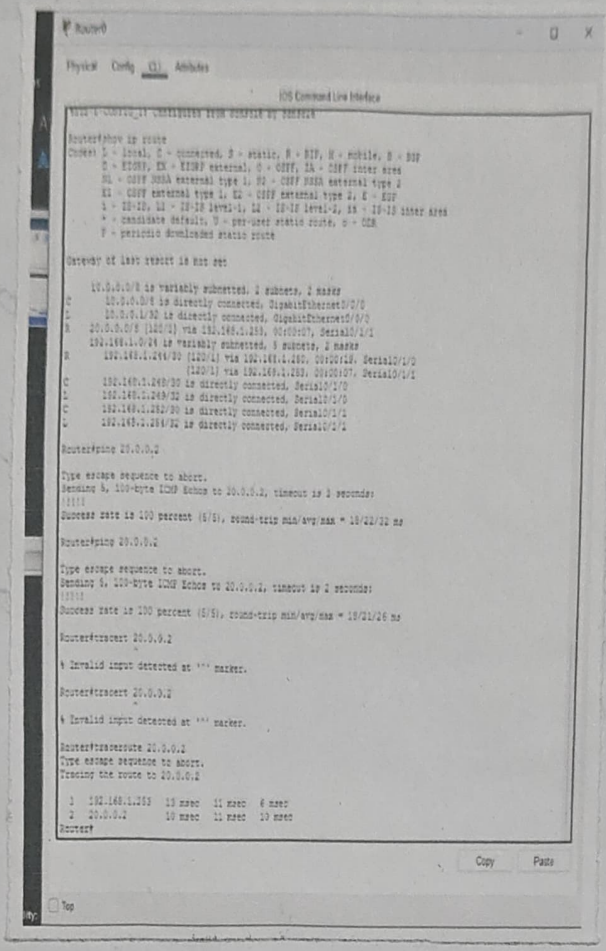
6) Enable RIP on Router 0 and configure RIP on Router 1 and 2.

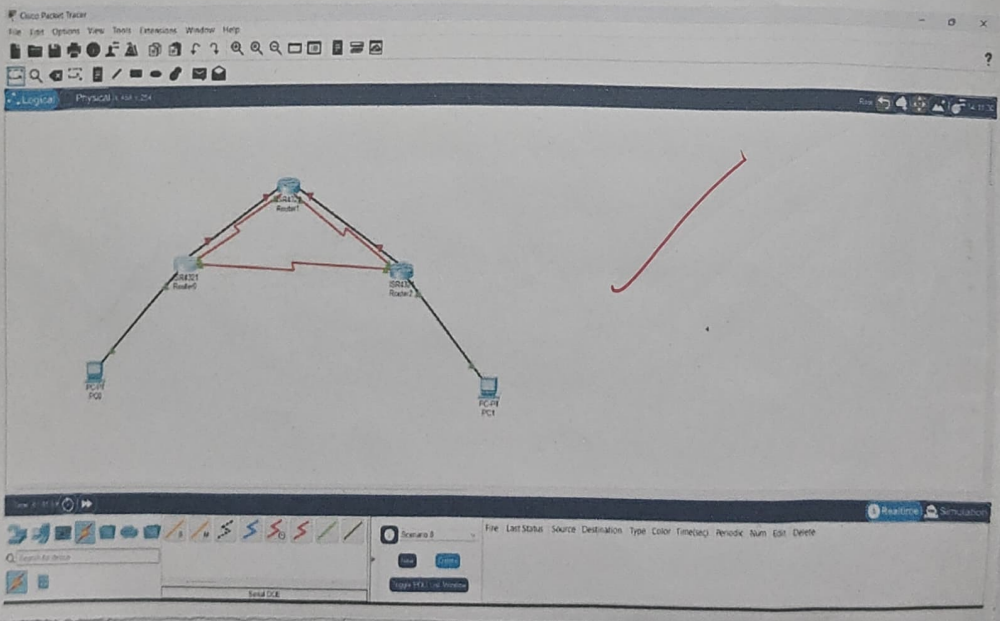
7) From PC0, use ping 20.0.0.2 to verify connectivity.

8) Use tracert to observe the route taken by packets.

9) Disconnect the direct Router 0 - Router 2 link and observe the RIP reroutes the traffic through Router 1.




Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, GigabitEthernet0/0/0
L 10.0.0.1/32 is directly connected, GigabitEthernet0/0/0
R 20.0.0.0/8 [120/1] via 192.168.1.253, 00:00:07, Serial0/1/1
192.168.1.0/24 is variably subnetted, 5 subnets, 2 masks
R 192.168.1.248/30 [120/1] via 192.168.1.250, 00:00:18, Serial0/1/0
[120/1] via 192.168.1.253, 00:00:07, Serial0/1/1
C 192.168.1.248/30 is directly connected, Serial0/1/0
L 192.168.1.249/32 is directly connected, Serial0/1/0
C 192.168.1.252/30 is directly connected, Serial0/1/1
L 192.168.1.254/32 is directly connected, Serial0/1/1
Router#ping 20.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/22/32 ms
Router#ping 20.0.0.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 20.0.0.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/21/26 ms
Router#traceort 20.0.0.2
% Invalid input detected at '^' marker.
Router#traceort 20.0.0.2
% Invalid input detected at '^' marker.
Router#traceroute 20.0.0.2
Type escape sequence to abort.
Tracing the route to 20.0.0.2
1 192.168.1.253 13 msec 11 msec 6 msec
2 20.0.0.2 10 msec 11 msec 10 msec
Router#
Copy
Paste
Top


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Router1
Router0
Router2
PC-PT
PC0
PC-PT
PC1
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Serial DCE

Result : Simulation of RIP using Cisco packet tracer for a network is implemented successfully.